

Excellent. Let's make this **tight, universal, and implementable**—not vibes, not marketing. This is a **protocol**, the way traffic lights are a protocol.

Below is a **Universal WeUP Light Language Specification (WLLS v1.0)** designed to work across *all* WeUP wearables, venues, and cultures, with **zero text, zero screens, and minimal learning curve**.

WeUP Light Language Specification (WLLS)

Version: 1.0

Scope: All WeUP wearables, shared tokens, and ambient devices

Design Goal: Convey *state, permission, and phase* without identity, distraction, or app dependence

1. Foundational Principles (Non-Negotiable)

1. **State, not identity**
Light never reveals who you are—only what mode you're in.
 2. **Ambient, not demanding**
Lights must be readable in peripheral vision.
 3. **Contextual meaning**
Color meaning is *mode-relative*, not global.
 4. **Finite vocabulary**
Fewer signals → faster cultural adoption.
 5. **Fail-safe silence**
“Off” always means privacy or non-participation.
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2. Signal Dimensions (The Grammar of Light)

Every WeUP signal is composed of **four orthogonal dimensions**:

Signal = Color × Motion × Intensity × Rhythm

Each dimension has a constrained vocabulary.

3. Color Layer — MODE (What you are doing)

Color answers one question only:

“What mode is this person or object currently in?”

Core Color Set (Universal)

These colors are **never redefined**, only contextualized.

Color	Semantic Anchor	Never Means
White	Neutral / Observing	Active play
Green	Active participation	Waiting
Amber	Transitional / Queue	Emergency
Blue	Focus / Cognitive	Social openness
Purple	Ritual / Ceremonial	Casual
Red (Dim)	Private / Closed	Danger
Cyan	Socially open	Performing
No Light	Non-participating / Invisible	Error

Example:
Green in a pool hall = “playing”
Green in a garden = “active tending”
Same intent, different domain.

4. Motion Layer — ROLE (How you relate to others)

Motion communicates **role dynamics**, not status.

Motion Pattern	Meaning
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Solid	Stable role
Slow Pulse (3–5s)	Waiting / Ready
Breathing (sinusoidal)	Attuned / Flow
Rotational Sweep	Turn-based activity
Soft Flicker	Open to interaction
Sharp Pulse (1s)	Attention required
Fade Out	Session ending

Motion is always subtle—no strobing, no flashing.

5. Intensity Layer — PERMISSION (Can I engage?)

Brightness communicates **social permission**, not urgency.

Intensity	Meaning
Very Low	Do not engage
Low	Observe only
Medium	Conditional engagement
High (never harsh)	Engagement welcome

Intensity is capped by venue rules (e.g., libraries enforce low max).

6. Rhythm Layer — PHASE (Where in the process)

Rhythm answers:

“Is this starting, ongoing, or ending?”

Rhythm	Phase
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Steady	Ongoing
Rising	About to begin
Falling	Winding down
Double Pulse → Steady	Transition complete
Single Pulse → Off	Exit / Closure

Rhythm changes **only at state transitions**, never continuously.

7. Canonical Composite Signals (Examples)

Pool Hall

In Queue

- Color: Amber
- Motion: Slow pulse
- Intensity: Medium
- Rhythm: Steady

Actively Playing

- Color: Green
- Motion: Solid
- Intensity: Medium
- Rhythm: Steady

Private Money Game

- Color: Red (dim)
- Motion: Solid
- Intensity: Low
- Rhythm: Steady

Watching / Available

- Color: White
- Motion: Soft flicker
- Intensity: Low

- Rhythm: Steady
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Meditation Spot

Entering

- Color: Purple
- Motion: Breathing
- Intensity: Very low
- Rhythm: Rising

In Session

- Color: Purple
- Motion: Breathing
- Intensity: Very low
- Rhythm: Steady

Completion

- Color: Purple → White
 - Motion: Fade
 - Intensity: Falling
 - Rhythm: Falling
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8. Shared Object Signals (Tables, Doors, Tokens)

Objects use the **same language**, never special cases.

Table Ready

- White, solid, medium

Table Locked (Game Active)

- Green, solid, medium

Table Transition

- Amber, rotational, medium

This creates **environmental coherence**—people read the room instantly.

9. Privacy & Safety Guarantees

- No signal exposes identity
- No signal encodes biometrics
- No signal persists beyond context
- No signal requires interpretation history

If context is unknown, signal defaults to **White / Solid / Low**.

10. Hardware Implementation Constraints

To enforce consistency:

- Minimum: **1 RGB LED + 1 haptic motor**
- Max brightness limited per venue profile
- Refresh rate $\leq 2\text{Hz}$
- No text, icons, or scrolling allowed

Violation = non-WeUP compliant.

11. Learning Curve (Why This Works)

Humans already know this language.

- Traffic lights
- Elevator indicators
- Studio “On Air” signs
- Do Not Disturb lamps

WeUP simply **extends an ancient visual grammar into social space**.

12. The Quiet Power

This system replaces:

- verbal interruptions
- social guessing
- phone checking
- awkward boundary enforcement

With:

ambient, consent-respecting clarity

No feeds.

No profiles.

No dopamine.

Just legible presence.

If you want next, we can:

- formalize this as a **W3C-style spec**
- create a **reference firmware**
- or design a **cross-cultural validation matrix** to ensure colors remain readable worldwide

This is one of those rare moments where *less interface* produces *more order*.